Q21 Deeper Project Suggestions

1. **Interface-substitution experiment:** build one 3-component system (perception → router → controller) on a Gymnasium classic-control task and swap only the seam — differentiable embeddings vs. discretized feature vector vs. text labels — holding components fixed. Measure task performance, sensitivity of downstream components to upstream retraining, and fault-localization time. The deliverable is a quantitative latency/observability table like the note's Section 5.
2. **Compounding-error curve:** generate k-stage synthetic pipelines where each stage has controllable per-stage accuracy; measure end-to-end success as a function of k and per-stage correlation; empirically test when joint fine-tuning of the last stage alone recovers pipeline performance. Compare against the geometric-decay prediction from the note.
3. **Options vs. prompt-orchestration head-to-head:** on a gridworld with 4–6 reusable skills, implement (a) a tabular options learner with termination conditions [**?**, **?**] and (b) an LLM (or scripted stand-in) that selects among the same skills from natural-language task descriptions. Report learning samples to competence, transfer to a new task, and failure-diagnosability — three numbers the two paradigms disagree about most.
4. **Load-balancing ablation:** extend the module's MoE toy: add the Shazeer load-balancing auxiliary loss [**?**], vary expert count (2–16) and cluster overlap, and plot when routing collapses to one expert. Identify the smallest auxiliary-loss weight that prevents collapse, and document the failure without it.